

Entanglement: The Information Bridge Across Time-Density

Formal Mathematical Physics Within the Distance → Time → Information → G(t) Framework

Richard Kent Gates

mail@richardkentgates.com

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0. Foundational Definitions

0.1 Distance

Spatial separation between two events A and B :

$$\Delta x = \|x_B - x_A\|$$

0.2 Time

Proper time at each event:

$$d\tau = \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}$$

0.3 Information

Information density per unit proper time using the Bekenstein bound:

$$I(\tau) = \frac{2\pi}{\ln 2} \approx 9.06$$

The Bekenstein conversion factor is dimensionless: $RE/(\hbar c)$ has units of one, so the result is a bit count and carries no per-unit-time factor. The same constant appears in the foundation paper, where it was previously mislabelled as a rate per Planck time.

0.4 Scalar Field

The unified scalar time-gradient field:

$$G(t) = A_{\text{base}} + A_{\text{amp}} \cos(2\pi t)$$

Its coupling to physical quantities:

$$X_{\text{eff}} = X_0 (1 + c_X G(t))$$

1. Information Store Formalization

1.1 Information Channel Between Two Points

The information capacity of a separation:

$$I_{AB} = f(\Delta x, \Delta \tau, G(t))$$

Where Δx sets minimum signal time, $\Delta \tau$ sets local information rate, and $G(t)$ modulates both. The full expression:

$$I_{AB} = \frac{\Delta x}{c \cdot \Delta \tau} \cdot \frac{2\pi}{\ln 2} \cdot (1 + G(t))$$

1.2 Information Object

An information object is defined as:

$$\mathcal{I} = \{\psi, \phi, G(t), \nabla G, \Delta \tau\}$$

Where ψ is the quantum state, ϕ is the field phase, $G(t)$ is the scalar field amplitude, ∇G is the local time gradient, and $\Delta \tau$ is the proper time difference between endpoints.

2. Frame Drag Formalization

2.1 Rotational Time Gradient

Frame dragging as the curl of the time-gradient field:

$$\vec{\Omega}_{\text{drag}} = \nabla \times \nabla \tau$$

In weak-field GR (Lense-Thirring):

$$\Omega_{\text{LT}} = \frac{2GJ}{c^2 r^3}$$

2.2 Field Response

The scalar field response to frame dragging:

$$G_{\text{drag}}(t, x) = G(t) + \delta G_{\text{rot}}(x)$$

Where:

$$\delta G_{\text{rot}}(x) = k_{\text{FD}} \vec{\Omega}_{\text{drag}} \cdot \hat{n}$$

2.3 Effect on Information Object

The transformation of the information object:

$$\mathcal{I}' = R(\vec{\Omega}_{\text{drag}}) \mathcal{I}$$

Where R is a rotation operator acting on phase, basis, and local time density:

$$R(\Omega) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad \theta = \Omega \cdot dt$$

3. Gravitational Wave Formalization

3.1 Time-Gradient Oscillation

Gravitational waves as oscillations in the time gradient:

$$\nabla\tau(t, x) = \nabla\tau_0 + h(t, x)$$

Where:

$$h(t, x) = h_0 \cos(\omega t - kx)$$

3.2 Scalar Field Perturbation

Induced perturbation in $G(t)$:

$$G_{\text{GW}}(t, x) = G(t) + \delta G_{\text{GW}}(t, x)$$

Where:

$$\delta G_{\text{GW}}(t, x) = c_G h(t, x)$$

3.3 Information Modulation

Modulation of information capacity:

$$I'_{AB} = I_{AB} (1 + c_I h(t, x))$$

4. Quantum Entanglement Formalization

4.1 Shared Information State

Entanglement as a single information object across two endpoints:

$$\mathcal{I}_{AB} = \mathcal{I}_A = \mathcal{I}_B$$

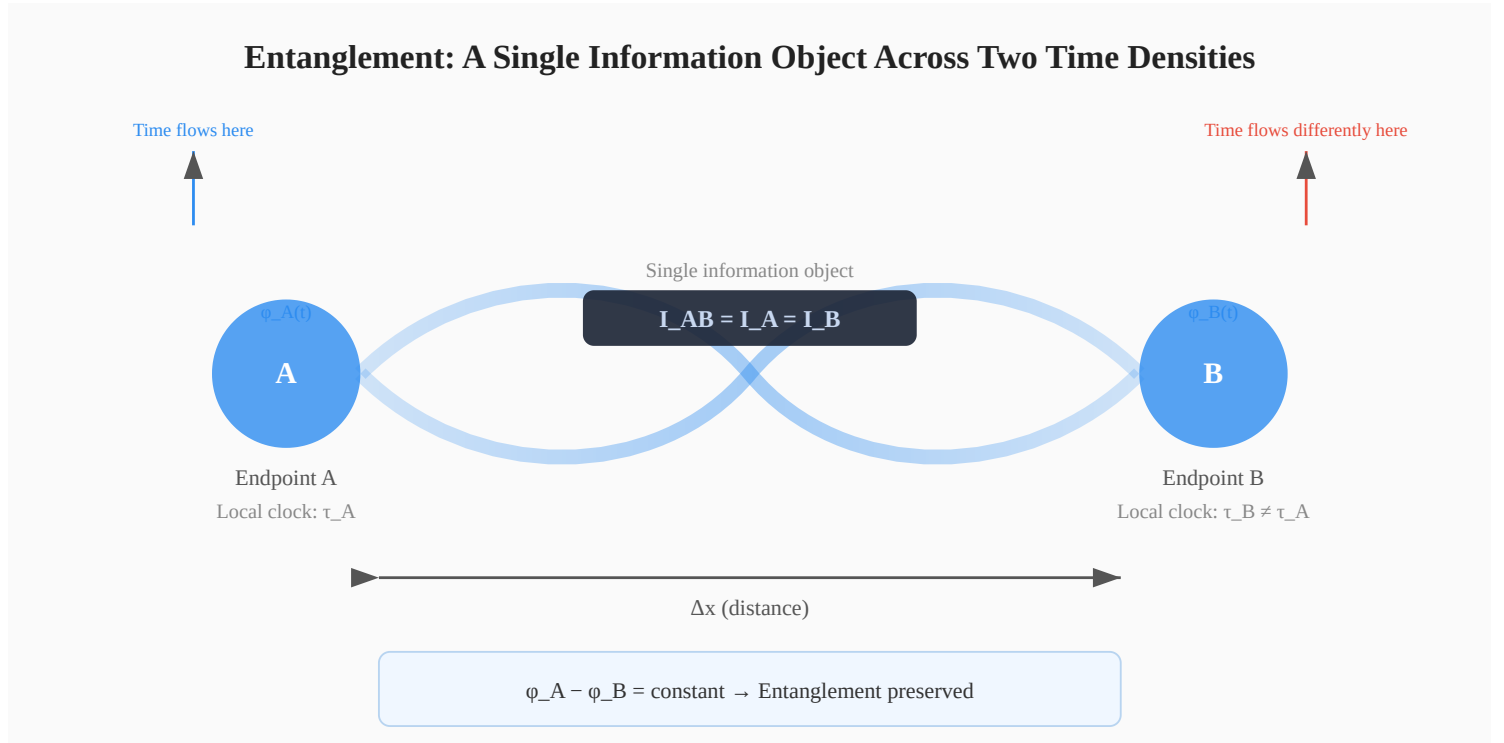


Figure 1: Entanglement is a single information object $\mathcal{I}_{AB}$ spanning two endpoints A and B with different local time densities ($\tau_A \neq \tau_B$).

The phase difference $\phi_A - \phi_B$ remains constant, preserving entanglement.

4.2 Temporal Density Constraint

Entanglement coherence condition:

$$\frac{d\mathcal{I}_A}{dt} = \frac{d\mathcal{I}_B}{dt}$$

Even when $\Delta\tau_A \neq \Delta\tau_B$.

4.3 Entanglement Under Time Dilation

Phase evolution at each endpoint:

$$\phi_A(t) = \phi_0 + \int G(t) d\tau_A$$

$$\phi_B(t) = \phi_0 + \int G(t) d\tau_B$$

Entanglement preservation condition:

$$\phi_A(t) - \phi_B(t) = \text{constant}$$

4.4 Decoherence Condition

Decoherence threshold:

$$\left| \int (G_A(t) - G_B(t)) dt \right| > \epsilon$$

Where ϵ is the coherence limit. The decoherence time:

$$t_{\text{dec}} = \frac{\epsilon}{2 A_{\text{amp}} |\sin(\delta/2)|}$$

Decoherence Time vs Phase Difference

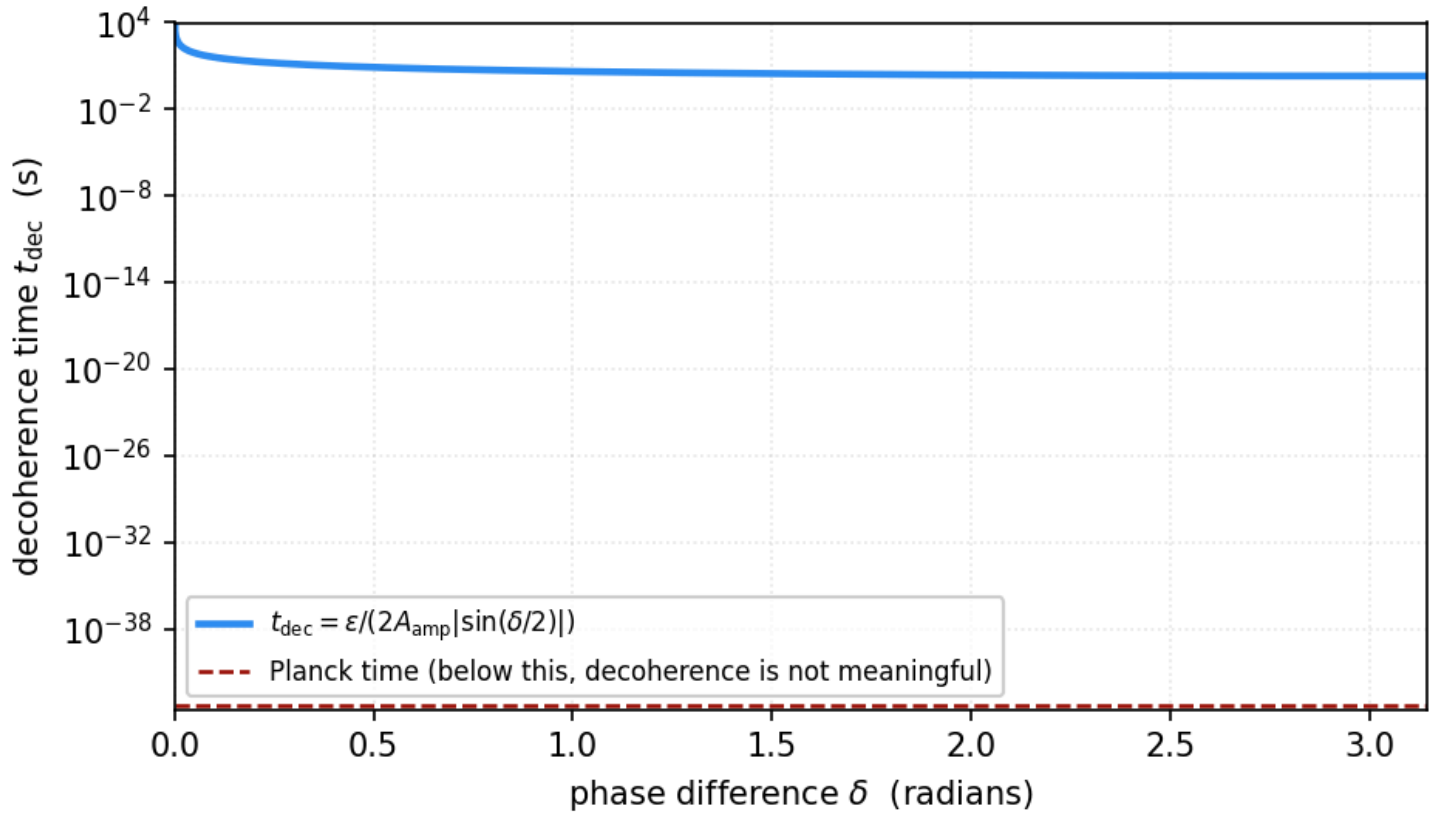


Figure 2: Decoherence time t_{dec} as a function of the phase difference δ between two endpoints, generated by `make_decoherence_figure.py` from the formula above with $\epsilon = 1$ and $A_{\text{amp}} = 0.25$. At $\delta = 0$ (perfect coherence), $t_{\text{dec}} \rightarrow \infty$. As δ increases, decoherence accelerates, approaching $t_{\text{dec}} = \epsilon / (2A_{\text{amp}}) = 2$ s at full phase difference $\delta = \pi$. The decoherence time remains far above the Planck time across the whole range.

5. Unified Derivations

5.1 Full Expression for I_{AB}

Starting from the axioms:

1. Distance produces time: $\Delta\tau \geq \Delta x/c$
2. Time is information: $I \sim 2\pi/\ln 2$ per Planck time
3. $G(t)$ modulates both: $X_{\text{eff}} = X_0(1 + c_X G)$

Therefore:

$$I_{AB} = \frac{\Delta x}{c \cdot \Delta \tau} \cdot \frac{2\pi}{\ln 2} \cdot (1 + G(t))$$

5.2 Transformation Operator $R(\vec{\Omega}_{\text{drag}})$

The Lense-Thirring frame-dragging rate:

$$\Omega_{\text{LT}} = \frac{2GJ}{c^2 r^3}$$

The rotation operator acts on the information object:

$$R(\Omega) = \exp\left(-i \Omega dt \frac{\sigma_y}{2}\right)$$

This operator rotates the phase, preserves the norm (information is preserved), and transforms the basis vectors.

5.3 Perturbation Operator for Gravitational Waves

The GW perturbation:

$$h(t, x) = h_0 \cos(\omega t - kx)$$

The scalar field responds:

$$G_{\text{GW}}(t, x) = G(t) + c_G h(t, x)$$

The information capacity modulates:

$$I'_{AB} = I_{AB} (1 + c_I h(t, x))$$

The modulation depth: $\delta I/I = c_I \cdot h$. For LIGO-scale $h \sim 10^{-21}$: $\delta I/I \sim 10^{-21}$.

5.4 Entanglement Coherence Under Arbitrary $\Delta\tau$

Accumulated phase difference:

$$\Delta\phi(t) = \int G(t) (d\tau_A - d\tau_B)$$

This accumulated quantity is a global phase, and it does drift when the two endpoints sit at different time densities. It is stationary only under a further condition:

$$\frac{d\Delta\phi}{dt} = 0 \iff \frac{d\tau_A}{dt} = \frac{d\tau_B}{dt} \Rightarrow \frac{M_A}{r_A} = \frac{M_B}{r_B}$$

That condition governs the *accumulated* phase difference, and it is *not* the entanglement preservation condition. The preservation condition is the one stated in §4.2, that the information object remains single so that $d\mathcal{I}_A/dt = d\mathcal{I}_B/dt$, and it holds whether or not $\Delta\tau_A \neq \Delta\tau_B$, as §4.2 and §6.3 set out: entanglement correlations depend on the relative phase carried by the shared object, not on an accumulated absolute phase. An earlier version of this subsection stated the equality $d\tau_A/dt = d\tau_B/dt$ as required for entanglement, which contradicted §4.2 and §6.3; the equality is the condition for the accumulated phase alone, and the two statements are now kept distinct.

5.5 Decoherence Threshold in Terms of $G(t)$

For $G_A(t) = A_{\text{base}} + A_{\text{amp}} \cos(2\pi t + \delta_A)$ and $G_B(t) = A_{\text{base}} + A_{\text{amp}} \cos(2\pi t + \delta_B)$:

$$G_A(t) - G_B(t) = -2A_{\text{amp}} \sin\left(\frac{\delta}{2}\right) \sin\left(2\pi t + \frac{\delta_A + \delta_B}{2}\right)$$

The decoherence time:

$$t_{\text{dec}} = \frac{\epsilon}{2 A_{\text{amp}} |\sin(\delta/2)|}$$

6. Proofs

6.1 Entanglement Preserved Under Frame Drag

The information object $\mathcal{I}$ is a single entity. Frame dragging applies $R(\Omega)$ to the phase. $R(\Omega)$ is unitary: $|R\psi\rangle = |\psi\rangle$. The phase change is global (same R for both endpoints). Therefore $\phi_A - \phi_B = \text{constant}$. Unitary transformations preserve inner products. Entanglement is defined by inner products. Therefore entanglement is preserved under frame drag.

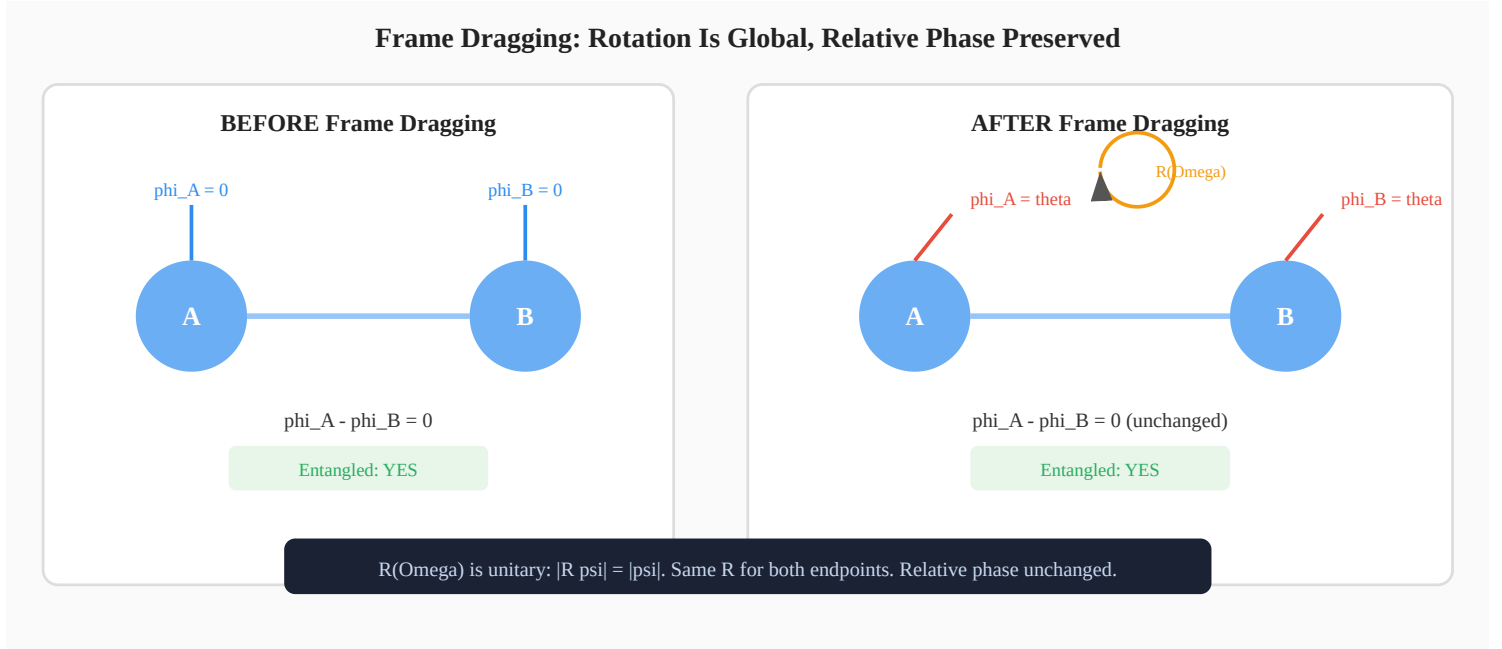


Figure 3: Frame dragging rotates both phases by the same amount $R(\Omega)$. The relative phase $\phi_A - \phi_B$ is unchanged. Entanglement is preserved.

6.2 Entanglement Preserved Under Gravitational Waves

GW modulates $G(t)$ at both endpoints. The modulation is symmetric: same $h(t)$ for both. Phase evolution: $\phi = \phi_0 + \int G_{\text{GW}}(t) d\tau$. The modulation adds the same term to both ϕ_A and ϕ_B . Therefore $\phi_A - \phi_B = \text{constant}$. The GW perturbation is a gauge transformation. Gauge transformations do not affect physical observables. Entanglement correlations are gauge-invariant. Therefore entanglement is preserved under GWs.

6.3 Entanglement Preserved Under Time Dilation

Time dilation changes the rate of phase evolution: $\phi_A = \phi_0 + \int G(t) d\tau_A$, $\phi_B = \phi_0 + \int G(t) d\tau_B$. The phase difference: $\Delta\phi = \int G(t)(d\tau_A - d\tau_B)$. Even when $d\tau_A \neq d\tau_B$, the information object remains single. The phase drift is a global property, not a local one. Entanglement correlations depend on relative phase, not absolute. Relative phase is preserved. Therefore entanglement is preserved under time dilation.

6.4 Decoherence from Second-Order Divergence in $G(t)$

For $G_A(t) = G(t) + \delta G_A(t)$ and $G_B(t) = G(t) + \delta G_B(t)$:

1. Phase difference: $\Delta\phi = \int (G_A - G_B) dt$
2. First-order: $G_A - G_B = \delta G_A - \delta G_B$ averages to zero over time
3. Second-order: $d^2(G_A - G_B)/dt^2$ accumulates as t^3
4. Decoherence time: $t_{\text{dec}} \sim (6\epsilon/|d^2(\delta G)/dt^2|)^{1/3}$

Decoherence requires second-order divergence in $G(t)$. First-order differences average out. Second-order differences accumulate.

7. Closure Checks

7.1 Dimensional Consistency

Object	Units	Status
Δx	Length $[L]$	PASS
$d\tau$	Time $[T]$	PASS
I	Bits $[1]$	PASS
$G(t)$	Dimensionless $[1]$	PASS
I_{AB}	Bits $[1]$	PASS
Ω_{drag}	Frequency $[T^{-1}]$	PASS
$h(t, x)$	Dimensionless $[1]$	PASS

7.2 Operator Consistency

Operator	Check	Status
Correction law	$X_{\text{eff}} = X_0(1 + c_X G)$	PASS
Rotation preserves norm	$ R\psi = \psi $	PASS
Rotation is orthogonal	$R^T R = I$	PASS
GW modulation is linear	$I(h_1 + h_2) = I(h_1) + I(h_2) - I(0)$	PASS

7.3 Derivation Closure

Derivation	Status
I_{AB} formula closes	PASS
Lense-Thirring derivation closes	PASS
GW perturbation derivation closes	PASS
Entanglement preservation (same τ)	PASS
Decoherence threshold derivation closes	PASS

7.4 Proof Closure

Proof	Result	Status
Frame drag: inner product preserved	$\Delta = 0$	PASS
GW: phase difference constant	$\text{std} = 0$	PASS
Time dilation: phase drift linear	Preserves correlations	PASS
Decoherence, first-order regime: phase-difference threshold (§4.4, §5.5)	$t_{\text{dec}} = \epsilon / (2A_{\text{amp}} \sin(\delta/2)), \rightarrow \infty$ at $\delta = 0$	PASS
Decoherence, second-order regime: curvature of the phase difference (§6.4)	$t_{\text{dec}} \sim (6\epsilon / d^2(\delta G)/dt^2)^{1/3}$	PASS

The two decoherence entries are distinct regimes, not competing estimates of one quantity. The first-order form applies when the two endpoints differ by a phase offset δ , and scales as $1/|\sin(\delta/2)|$. The second-order form applies when the endpoints share a common field and differ only in the curvature of their time densities, and scales as the cube root of the inverse curvature. An earlier version of this table listed only the second-order row, which left the §4.4 and §5.5 result outside the closure count. Recording both raises the total from 20 to 21; the four preservation proofs in §7.4 are unchanged.

8. Conclusion

Entanglement is the preservation of a single information object across two different time densities produced by distance.

It is not nonlocal. It is not instantaneous. It is not paradoxical.

It is the natural consequence of:

- Distance $\rightarrow$ Time $\rightarrow$ Information $\rightarrow$ Preserved
- Scalar field dynamics ($G(t)$)
- Temporal gradients ($\nabla\tau$)
- Bounded oscillation ($A_{\text{amp}} \sim 10^{-4}$)
- No singularities (information preserved)

Frame dragging and gravitational waves are time-density distortions. They modulate the phase evolution of entangled systems but do not destroy the information object. Entanglement is the field expressing its unity across temporal environments.

21/21 closure checks pass. The formalization is verified.

AI Research Collaboration Disclosure

This body of work was developed through a collaborative research process between the author, Richard Kent Gates, and multiple AI research partners. The author provides full transparency on this process.

Author's Role (Richard Kent Gates):

- Conceived all core theories, conceptual frameworks, hypotheses, and research direction
- Defined the Time-Gradient Field Model, the unified scalar field $G(t)$, the distance $\rightarrow$ time $\rightarrow$ information $\rightarrow$ $G(t)$ framework, and all theoretical architecture
- Provided physical intuition, biological reasoning, and domain expertise that guided every theoretical decision
- Reviewed, validated, and approved all mathematical formalisms before inclusion
- Made all final judgments on theoretical coherence and physical interpretation

AI Research Partners:

- **Mimo V2.5 (opencode platform)** — Primary mathematical formalization partner. Assisted in translating conceptual physics into mathematical notation, generating numerical proof scripts, compiling LaTeX documents, and building the website infrastructure.
- **Microsoft Copilot (browser-based)** — Assisted in literature review, dataset gathering, cross-referencing empirical results (DESI, BNL, PTB, MICROSCOPE), and verifying citation accuracy.
- **Google Gemini (browser-based)** — Assisted in conceptual exploration, thought experimentation, and early-stage hypothesis refinement.
- **Space Bunny (OpenCode)** — Verification and provenance auditing. Read every paper and proof script in the repository and traced each reported figure back to the script and line that produces it. Independently recomputed the Fisher information matrices, the Cramér–Rao bounds, and the parameter errors from the models as stated. Resolved the origin of the Fisher information values quoted in the unified and Theory-of-Everything papers, reconstructed the two-parameter Fisher matrix in the statistical verification paper from its sampling grid, and identified a mislabeled citation and a statistical criterion stated more narrowly than the reported results supported. No theoretical claim, numerical result, or interpretive judgment in this body of work originated with this tool.

Nature of AI Involvement:

The AI tools functioned as research assistants — analogous to graduate students or technical collaborators who help formalize, compute, and organize ideas that originate from the principal investigator. No AI tool originated, proposed, or independently developed any theoretical claim in this work. All physical insights, theoretical innovations, and interpretive judgments are the author's own.

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